

OpenShift Container Platform - Course Outline

1. Duration

- 48 Hours (12 Half-Day Sessions)

2. Objectives

At the end of this workshop, participants will be able to:

- **Understand containerization principles and Kubernetes fundamentals**, including core concepts, architecture, and cloud-native application deployment models.
- **Understand OpenShift architecture, components, and product capabilities**, and how OpenShift extends Kubernetes into an enterprise container platform.
- **Administer and operate OpenShift clusters**, including cluster lifecycle management, node operations, and platform administration tasks.
- **Configure and manage core platform services**, including networking, ingress and routes, storage, security, authentication, monitoring, and logging.
- **Deploy and manage applications on OpenShift** using Kubernetes manifests, Helm charts, Operators, and GitOps-based deployment workflows.
- **Monitor, troubleshoot, and maintain OpenShift environments**, including backup and recovery, platform health checks, and diagnosing cluster and application issues.

3. Audience

- Openshift Administrators, Linux Administrators, Infrastructure Engineers
- Platform Engineers, DevOps Engineers, Cloud Administrators
- Operations Engineers, Application Administrators

4. Prerequisites

- Linux Administration Fundamentals
- Basic Networking Concepts
- Basic Container Knowledge
- Basic Kubernetes Awareness

5. Hardware & Software Requirements

- Desktop/Laptop with minimum 16 GB RAM
- Open Internet connection (minimum 10 Mbps per user)
- Local Admin Access
- Git client, SSH Client
- kubectl, OpenShift CLI (oc), Helm
- VS Code
- Docker / Podman Desktop
- Browser (Chrome/Safari/Edge/Firefox)
- OpenShift Developer Sandbox Access
- Shared OpenShift 4.18 Cluster Access

6. Outline

Module 1: Container Fundamentals & Runtime Refresher

Topics

- Evolution of application deployment
- Virtual Machines vs Containers
- Container architecture and OCI concepts
- Images, registries, and container lifecycle
- Docker and Podman overview
- Container networking and storage fundamentals

Hands-on

- DockerHub account registration
- Run and manage containers
- Build container images
- Push and pull images from a registry

✓ **Objective Alignment:** Understand containerization principles and Kubernetes fundamentals, including core concepts, architecture, and cloud-native application deployment models.

Module 2: Kubernetes Foundations and Architecture

Topics

- Why Kubernetes
- Kubernetes architecture and components
- Control Plane and Worker Nodes
- API Server
- etcd
- Scheduler
- Controller Manager
- Kubelet
- Container Runtime
- Kube Proxy
- Pods
- Declarative configuration using YAML

Hands-on

- kubectl CLI commands walkthrough
- Kubernetes Object Exploration
- Namespace Management,
- Kubernetes Dashboard access and exploration
- Creating and access the pod

✓ **Objective Alignment:** Understand containerization principles and Kubernetes fundamentals, including core concepts, architecture, and cloud-native application deployment models.

Module 3: Kubernetes Workloads, Storage and Networking

Topics

- Pods, ReplicaSets, Deployments
- StatefulSets, DaemonSets, Jobs
- Services and Namespaces
- ConfigMaps and Secrets
- Volumes and Networking fundamentals
- Ingress controller overview

Hands-on

- Application Deployment
- Service Exposure
- Rolling Updates & Rollbacks
- Volumes and Networking

✓ **Objective Alignment:** Understand containerization principles and Kubernetes fundamentals, including core concepts, architecture, and cloud-native application deployment models.

Module 4 — OpenShift Architecture and Product Overview

Topics

- OpenShift evolution and release history
- OpenShift stack architecture
- OpenShift product portfolio
- OpenShift vs Kubernetes
- RHCOS and CRI-O
- Operators and Cluster Operators
- Machine API

Hands-on

- OpenShift Console walkthrough
- oc CLI essentials
- Cluster architecture walkthrough

✓ **Objective Alignment:** Understand OpenShift architecture, components, and product capabilities, and how OpenShift extends Kubernetes into an enterprise container platform.

Module 5 — OpenShift Installation and Cluster Administration

Topics

- Cluster topology — Control Plane, Worker, and Infrastructure nodes
- Cluster Operators and the Machine Config Operator
- Cluster Version Operator
- Installation models overview
- Cluster lifecycle management
- Managing projects and users

Hands-on

- Explore cluster operators
- Verify cluster health
- Perform administrative operations
- Manage projects and users

✓ **Objective Alignment:** Administer and operate OpenShift clusters, including cluster lifecycle management, node operations, and platform administration tasks.

Module 6 — OpenShift Networking, Routes and Multi-Tenancy

Topics

- OpenShift networking architecture and OVN-Kubernetes overview
- Services and DNS in OpenShift
- OpenShift Routes: Edge, Passthrough, and Re-encrypt termination
- Routes vs Kubernetes Ingress
- Network Policies fundamentals
- Multi-tenancy patterns for shared clusters

Hands-on

- Create and manage Routes
- Configure TLS termination
- Apply Network Policies for tenant isolation
- Expose applications externally

✓ **Objective Alignment:** Configure and manage core platform services, including networking, ingress and routes, storage, security, authentication, monitoring, and logging.

Module 7 — OpenShift Storage and Security Fundamentals

Storage

- Persistent Volumes and Persistent Volume Claims
- Storage Classes and dynamic provisioning
- OpenShift Data Foundation overview

Security

- Authentication and Authorization fundamentals
- Security Context Constraints (SCC)
- Service Accounts and Secrets management

Hands-on

- Configure Storage Classes and provision persistent storage
- Apply Security Context Constraints
- Manage Secrets and Service Accounts

✓ **Objective Alignment:** Configure and manage core platform services, including networking, ingress and routes, storage, security, authentication, monitoring, and logging.

Module 8 — OpenShift Authentication, RBAC and Identity Integration

Topics

- RBAC concepts — Roles, RoleBindings, ClusterRoles
- Designing least-privilege access models
- LDAP integration overview
- Active Directory integration overview
- OAuth configuration
- OpenID Connect overview

Hands-on

- Create custom Roles and RoleBindings
- Configure an external identity provider
- Validate access with test users and groups

✓ **Objective Alignment:** Configure and manage core platform services, including networking, ingress and routes, storage, security, authentication, monitoring, and logging.

Module 9 — Operators, Helm and Application Deployment

Topics

- Operator Framework and Operator Lifecycle Manager (OLM)
- OperatorHub
- Kubernetes manifests for application deployment
- Helm fundamentals — charts, values, releases

Hands-on

- Install and manage Operators via OLM
- Deploy applications using YAML manifests
- Deploy and customize applications using Helm

✓ **Objective Alignment:** Deploy and manage applications on OpenShift using Kubernetes manifests, Helm charts, Operators, and GitOps-based deployment workflows.

Module 10 — GitOps, CI/CD and Service Mesh

GitOps and CI/CD

- GitOps principles
- OpenShift GitOps (Argo CD) architecture
- Jenkins integration and pipeline concepts
- Jenkins to GitOps deployment flow

Service Mesh

- Service Mesh fundamentals and use cases
- Sidecar proxy model and traffic management overview

Hands-on

- Configure Argo CD applications
- Demonstrate Jenkins to GitOps deployment flow
- Explore a basic Service Mesh deployment

✓ **Objective Alignment:** Deploy and manage applications on OpenShift using Kubernetes manifests, Helm charts, Operators, and GitOps-based deployment workflows.

Module 11 — Monitoring, Logging and etcd Administration

Topics

- Cluster monitoring architecture
- Prometheus and Alertmanager
- Grafana dashboards
- Logging architecture overview
- ETCD architecture, health, and performance monitoring

Hands-on

- Explore monitoring dashboards and create a custom alert rule
- Query and filter platform logs
- Verify ETCD health and performance metrics

✓ **Objective Alignment:** Monitor, troubleshoot, and maintain OpenShift environments, including backup and recovery, platform health checks, and diagnosing cluster and application issues.

Module 12 — Backup, Recovery, Troubleshooting and Capstone

Backup, Recovery and Node Lifecycle

- Backup strategies and ETCD backup and recovery
- Worker node replacement and control plane replacement overview

Troubleshooting

- Cluster and platform troubleshooting methodology
- Application troubleshooting

Capstone Lab

- End-to-end deployment exercise integrating networking, storage, security, and GitOps
- Guided review and Q&A

Hands-on

- ETCD backup and restore demonstration
- Replace a failed worker node
- Troubleshoot application and cluster operator failures
- Capstone: deploy a multi-tier application with Ingress/Route, PVC, Secrets, and an Argo CD-managed pipeline

✓ **Objective Alignment:** Monitor, troubleshoot, and maintain OpenShift environments, including backup and recovery, platform health checks, and diagnosing cluster and application issues.